

Fall 2026

MSCS 435/635 – Advanced Machine Learning

TTH 9:40 AM – 11:05 AM (JHSW 103)

Instructor: Dr. Augustine Twumasi

Office: JHSW 234G

What to Call me: Please call me Dr. Twumasi

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Email: twumasia@uwstout.edu

Student Hours (Office Hours)

Student hours (also known as office hours) are set aside for discussing course material, assignments, projects, or any questions related to the class. All students are welcome and strongly encouraged to attend. You do not need to notify me in advance if you plan to stop by during the scheduled office hours.

If you are unable to attend during the listed times and would like to meet outside of office hours, please email me in advance and include a few possible times that work for you. I will do my best to coordinate a meeting that fits both our schedules.

Office Hours: In my office or through Teams

- MTWTHW: 11:10 – 12:10
- TR: 12:20 – 1:15
- F: 10:10 – 11:10
- By Appointment

COURSE DESCRIPTION

Deep learning systems are increasingly central to modern AI, powering applications from generative models to language understanding and autonomous decision-making. This course covers modern machine learning architectures, with hands-on applications to computer vision, generative modeling, sequence modeling, and natural language processing. Topics include convolutional neural networks, U-Net, and residual neural networks, generative architectures including variational autoencoders, generative adversarial networks, and diffusion models, recurrent neural networks, long short-term memory, and transformers, word embeddings, large language models, and retrieval-augmented generation, and reinforcement learning including multi-armed bandits and deep Q-learning, tree search algorithms and their combination with neural networks (as in AlphaGo). By the end of the course, students will be able to implement these architectures directly and will be well positioned to understand and extend their knowledge through current research literature.

COURSE OBJECTIVES:

1. Upon successful completion of the course, the student will be able to
2. Use reinforcement learning to train an AI to play a game.
3. Create an AI that determines the optimal or near-optimal move in a game.
4. Understand the role of convolutions in feature extraction.
5. Create image classification systems.
6. Use image segmentation for feature detection and extraction.
7. Understand multiple generative AI architectures.
8. Implement a word embedding.
9. Explain the significance of the model constraints of a large language model, including context window size, token limit, training scope, and temperature.
10. Augment large language models with a custom knowledge base.

Graduate students should also be able to:

11. Create a neural network given its architecture diagram.

COURSE CONTENT

1. **Foundations of Deep Learning**
 - a. Neural networks and backpropagation
 - b. Loss functions and optimization
 - c. Gradient descent and learning-rate strategies
 - d. Regularization and generalization
 - e. Training, validation, and testing
 - f. Hyperparameter tuning
2. **Computer Vision**
 - a. Convolutional neural networks
 - b. Residual neural networks
 - c. Image classification
 - d. Object detection
 - e. Image segmentation
 - f. Special architectures: Siamese networks, metric learning (ArcFace), and Vision Transformers
3. **Generative Modeling**
 - a. Latent representations and generative modeling
 - b. Variational autoencoders
 - c. Generative adversarial networks
 - d. Diffusion models
 - e. Conditional generation and latent spaces
 - f. Special architectures: StyleGAN, CycleGAN, and PatchGAN

4. **Sequence and Attention Modeling**
 - a. Recurrent neural networks
 - b. Long short-term memory networks
 - c. Attention mechanisms
 - d. Transformer architectures
 - e. Encoder, decoder, and encoder-decoder architectures
 - f. Special architectures: GRUs, and BERT vs. GPT (encoder-only vs. decoder-only)
5. **Natural Language Processing and Foundation Models**
 - a. Word and contextual embeddings
 - b. Language modeling and pretraining
 - c. Large language models
 - d. Fine-tuning and parameter-efficient fine-tuning
 - e. Prompt engineering
 - f. Retrieval-augmented generation
 - g. Special Topics: Knowledge Distillation and Model Compression
6. **Reinforcement Learning and Decision Making**
 - a. Markov decision processes
 - b. Multi-armed bandits
 - c. Q-learning
 - d. Deep Q-networks
 - e. Policy-based reinforcement learning
 - f. Monte Carlo Tree Search
 - g. AlphaGo and deep reinforcement learning
 - h. Special architectures: MuZero

PREREQUISITES

MSCS 335 Machine Learning and

PROGRAMMING PLATFORM

We will use Google Colab for coding exercises and assignments.

GRADING

- Class activities and participation - 10%
- Paper homework - 10%
- Midterm Exam - 15%

- Projects – 60%

Policy on late assignments:

Project and homework grades will be reduced by a factor of 25% for each working day or fraction they are late.

Attendance and Participation

This grade (10% of the final grade) reflects regular attendance and active participation. Participation includes contributing to class discussions, engaging during in-class activities, and being prepared for lectures. It reflects consistent, meaningful engagement rather than attendance alone, and supports learning and collaborative inquiry.

Paper Homework

Short, on-paper problem sets that test conceptual and mathematical understanding independent of code. Sets on Foundations and Computer Vision are graded and serve as direct preparation for the midterm exam; sets released after the midterm are lighter, completion-based practice tied to ongoing project work.

Midterm Exam

Comprehensive, in-person final on **Thursday, May 14, 12:00–1:50 PM**. Strong performance is expected with proper preparation throughout the semester. The exam will be in person, and you are expected to be here.

The exam covers the Foundations of Deep Learning and Computer Vision units. It emphasizes conceptual clarity, problem-solving ability, and the ability to reason about model behavior rather than memorization.

Projects

Rather than a single project due at the end of the semester, students complete three individual, applied machine learning projects spaced across the course so that implementation work builds progressively alongside lecture content. Students choose their own project topic for each round, applying the architectures and techniques covered in the units completed up to that point. Each project includes a short one-on-one checkpoint review with the instructor and culminates in a brief write-up and demonstration.

Grading Scale: Grades will be based on the following scale:

Points	Grade
92-100	A
90-91.99	A-
88—89.99	B+
82-87.99	B
80-81.99	B-
78-79.99	C+
72-77.99	C
70-71.99	C-
68-69.99	D+
62-67.99	D
60-61.99	D-
0-59.99	F

Syllabus Change Policy: The instructor reserves the right to make changes throughout the semester to the syllabus and schedule if necessary. It is each student’s responsibility to be present and regularly review the course management system for updates.

TEXTBOOK

There will be no textbook. We will use parts of online books and other materials.

Email Communication Policy

Email is the preferred method for contacting me with course-related questions. During regular business hours, I will respond as promptly as possible. Emails sent after **5:00 PM on weekdays or on Sundays** will typically be answered **the next business day**, and emails sent on **Saturdays will be responded to on Monday**.

To help me respond efficiently, please follow these guidelines when emailing:

- Use the subject line format:
“MSCS 335-001: [Brief description of your email]”

- Begin your message with a professional salutation, such as:
“Dear Dr. Twumasi” or “Hello Dr. Twumasi.”
- Write in complete sentences and clearly state your question or concern.
- End your email with a courteous closing, such as:
“Best regards,” or “Sincerely,” followed by your full name.

Canvas Information: The Learning Management System for this course will be Canvas. This can be found at [canvas-link](#) . You can login with your UW-Stout username and password.

Important dates/deadlines: For important dates and deadlines refer to [university academic calendar](#)

University Policies: It is expected that students follow University policies that include, but are not limited to, academic misconduct, disability accommodations, sexual harassment, non-discrimination, etc. Please see the university website regarding policies that affect all classrooms on campus at the following link: [University Policies](#).

In particular, if you qualify for any disability accommodations, please contact me and provide the appropriate paperwork so that we can get those accommodations provided as soon as possible.

Academic Honesty: One of the University Policies included in the link above deals with academic honesty and academic misconduct. It is important that students are responsible for the honest completion of their work. This includes citing any resources used and doing one's own work. Cases of academic dishonesty tend to come about when students panic about an upcoming assignment or exam. Please start your work and ask questions early to avoid getting into that type of situation. Also, please reach out to me if you are at all unclear about what academic dishonesty entails and how to avoid it. If I discover an incident of academic dishonesty, I will discuss the issue with the student(s) in question. Consequences will vary depending on the severity of the situation. For more information, see [Stout Academic Misconduct Policy](#) and [Stout's Academic Investigation Process](#).

Course Concerns: If you have questions or concerns regarding this course, please come talk to me. As mentioned earlier, I want to help guide you as you learn Machine Learning. I hope you feel comfortable coming to me with questions and concerns, but if you would like to speak to someone else, according to the [Stout Complaint Policy](#), you can reach out to my department chair, Dr. Nelu Ghenciu, at (ghenciup@uwstout.edu)

Workload Expectations: As stated in the University Policy about credit hours, students are expected to spend at minimum twice as much time per week working on this course outside of class as time spent in class. This class is a 3-credit hour class, so you are expected to spend 3 hours in class per week and 6–9 hours per week working on this course outside of class. If you think you cannot commit to this workload, I recommend that you take this class at another time